

BASE PAPER — segmented gate driver, our reimplementation
Zhang, Yu, Leng, Cui, Deng, Ng. ISPSD 2020, pp. 102–105. 7 slices per bank, engaged in TWO stages.

STAGE 1 — engaged immediately

STAGE 2 — joins at the pattern step

VP (drive rail)

OUT (to HEMT gate)

VN (tied to ref — NO negative rail)

HOW THEIR CONTROL WORKS, AND WHY THIS SHEET IS WIDER THAN OURS.

The two stages are separate branches in the netlist, so 7 slices per bank means 14 columns. A stage–2 branch carries an EXTRA switch in series, gated by the pattern step (seg) rather than by the drive command. That extra switch is the paper’s mechanism: nseg slices engage at the edge and the remaining 5 join TSTEP later.

The split, and therefore the whole pattern, is set by ONE external bias resistor chosen at design time. That single knob is the paper’s contribution and is what ‘Simple Driving Strength Pattern Control’ in its title refers to.